

Performance Testing

Date	1 July 2026
Team ID	XXXXXX
Project Name	Smart Agricultural Production Optimization Engine (OptiCrop)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Web Browser
Type of Testing	Load Testing
Target Module / API	Crop Recommendation Prediction Module
Test Environment	Local System (Flask Development Server)
Test Date	1 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user submits crop prediction	1	30	Prediction displayed successfully
2	Multiple prediction requests	10	60	Stable performance without errors
3	Continuous prediction requests	20	120	Consistent response time
4	Invalid input values	5	30	Validation message displayed

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 sec	1.2 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	2.8 sec	Pass	Within limit
3	Throughput (Req/sec)	>5	8 Req/sec	Pass	Stable throughput
4	Error Rate	< 1%	0%	Pass	No errors observed
5	CPU Utilization	< 80%	42%	Pass	Normal CPU usage
6	Memory Utilization	< 80%	42%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings:

[Describe what the test results revealed about your system's performance]

Bottlenecks Identified:

[List any performance bottlenecks observed during testing]

Optimization Steps Taken:

[Describe any changes or optimizations made to improve performance]

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

